

Complete MySQL Date and Time Functionalities Guide

This guide covers:

- 1. Date and Time Data Types
- 2. Current Date/Time Functions
- 3. Date Extraction Functions
- 4. Date Formatting Functions
- 5. String to Date/Time Conversion
- 6. Type Casting and Conversion
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1. MySQL Date and Time Data Types

Data Type	Format	Description
DATE	YYYY-MM-DD	Stores only date
TIME	HH:MM:SS	Stores only time
DATETIME	YYYY-MM-DD HH:MM:SS	Stores date and time
TIMESTAMP	YYYY-MM-DD HH:MM:SS	Stores timestamp with timezone conversion
YEAR	YYYY	Stores year

2. Create Sample Table

```
CREATE TABLE employee_logs (
```

```
id INT PRIMARY KEY AUTO_INCREMENT,  
employee_name VARCHAR(100),  
joining_date DATE,  
login_time TIME,  
created_at DATETIME,  
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

3. Insert Sample Data

```
INSERT INTO employee_logs  
(employee_name, joining_date, login_time, created_at)  
VALUES  
( 'Arun Kumar', '2024-01-15', '09:30:00', '2024-01-15 09:30:00'),  
( 'Priya Sharma', '2023-07-20', '10:15:30', '2023-07-20 10:15:30'),  
( 'Vikram Reddy', '2022-12-01', '08:45:10', '2022-12-01 08:45:10');
```

4. Current Date and Time Functions

Function	Description
NOW()	Current date and time
CURRENT_TIMESTAMP()	Current timestamp
CURDATE()	Current date
CURRENT_DATE()	Current date
CURTIME()	Current time
CURRENT_TIME()	Current time
SYSDATE()	Current execution time

Examples

```
SELECT NOW();
```

```
SELECT CURRENT_TIMESTAMP();
```

```
SELECT CURDATE();
```

```
SELECT CURTIME();
```

```
SELECT SYSDATE();
```

Difference Between NOW() and SYSDATE()

```
SELECT NOW(), SLEEP(5), NOW();
```

```
SELECT SYSDATE(), SLEEP(5), SYSDATE();
```

Explanation:

- NOW() returns same time during query execution
 - SYSDATE() returns actual current execution time
-

5. Extracting Parts from Date

Function	Description
YEAR()	Extract year
MONTH()	Extract month
DAY()	Extract day
HOUR()	Extract hour
MINUTE()	Extract minute
SECOND()	Extract second
WEEK()	Week number
QUARTER()	Quarter
DAYNAME()	Day name

MONTHNAME() Month name

Examples

```
SELECT YEAR('2025-08-15');
```

```
SELECT MONTH('2025-08-15');
```

```
SELECT DAY('2025-08-15');
```

```
SELECT HOUR('2025-08-15 10:30:45');
```

```
SELECT MINUTE('2025-08-15 10:30:45');
```

```
SELECT SECOND('2025-08-15 10:30:45');
```

```
SELECT DAYNAME('2025-08-15');
```

```
SELECT MONTHNAME('2025-08-15');
```

```
SELECT QUARTER('2025-08-15');
```

6. DATE_FORMAT()

Used to format date/time.

Common Format Specifiers

Specifier	Meaning
%Y	4-digit year
%y	2-digit year
%M	Full month
%m	Numeric month

%d	Day
%W	Weekday
%H	Hour (24)
%h	Hour (12)
%i	Minutes
%s	Seconds
%p	AM/PM

Examples

```
SELECT DATE_FORMAT(NOW(), '%d-%m-%Y');
```

```
SELECT DATE_FORMAT(NOW(), '%W %M %Y');
```

```
SELECT DATE_FORMAT(NOW(), '%d/%m/%Y %h:%i:%s %p');
```

```
SELECT DATE_FORMAT('2025-08-15 14:30:45',  
    '%W, %d %M %Y %h:%i %p');
```

7. String to Date/Time Conversion

STR_TO_DATE()

Converts string into DATE/DATETIME.

Syntax

```
STR_TO_DATE(string, format)
```

Examples

Convert String to DATE

```
SELECT STR_TO_DATE('15-08-2025', '%d-%m-%Y');
```

Output:

2025-08-15

Convert String to DATETIME

```
SELECT STR_TO_DATE('15/08/2025 10:30:45',  
    '%d/%m/%Y %H:%i:%s');
```

Convert Month Name String

```
SELECT STR_TO_DATE('August 15 2025',  
    '%M %d %Y');
```

Insert String Date into Table

```
INSERT INTO employee_logs  
(employee_name, joining_date)  
VALUES  
(  
    'Rahul Verma',  
    STR_TO_DATE('25-12-2024', '%d-%m-%Y')  
);
```

8. CAST() and CONVERT()

CAST()

Used to convert datatype.

Syntax

CAST(value AS datatype)

CAST Examples

String to DATE

SELECT CAST('2025-08-15' AS DATE);

String to DATETIME

SELECT CAST('2025-08-15 10:20:30' AS DATETIME);

String to TIME

SELECT CAST('10:45:55' AS TIME);

DATETIME to DATE

SELECT CAST(NOW() AS DATE);

DATETIME to TIME

SELECT CAST(NOW() AS TIME);

String to YEAR

SELECT CAST('2025' AS YEAR);

CONVERT()

Alternative conversion function.

Syntax

CONVERT(value, datatype)

Examples

```
SELECT CONVERT('2025-08-15', DATE);
```

```
SELECT CONVERT('2025-08-15 11:30:45', DATETIME);
```

```
SELECT CONVERT(NOW(), DATE);
```

9. Date Arithmetic Functions

Function	Description
DATE_ADD()	Add interval
DATE_SUB()	Subtract interval
ADDDATE()	Add date
SUBDATE()	Subtract date

Add Days

```
SELECT DATE_ADD('2025-08-15', INTERVAL 10 DAY);
```

Add Months

```
SELECT DATE_ADD('2025-08-15', INTERVAL 2 MONTH);
```

Add Years

```
SELECT DATE_ADD('2025-08-15', INTERVAL 1 YEAR);
```

Add Hours

```
SELECT DATE_ADD(NOW(), INTERVAL 5 HOUR);
```

Subtract Days

```
SELECT DATE_SUB('2025-08-15', INTERVAL 7 DAY);
```

Supported INTERVAL Units

Unit

SECOND

MINUTE

HOURL

DAY

WEEK

MONTH

YEAR

10. Date Difference Functions

Function	Description
DATEDIFF()	Difference in days

TIMESTAMPDIFF() Difference in specified unit

DATEDIFF()

```
SELECT DATEDIFF('2025-08-15', '2025-08-01');
```

TIMESTAMPDIFF()

Difference in Days

```
SELECT TIMESTAMPDIFF(  
    DAY,  
    '2025-08-01',  
    '2025-08-15'  
);
```

Difference in Months

```
SELECT TIMESTAMPDIFF(  
    MONTH,  
    '2024-01-01',  
    '2025-08-01'  
);
```

Difference in Years

```
SELECT TIMESTAMPDIFF(  
    YEAR,  
    '2000-01-01',  
    CURDATE()  
);
```

11. Time Functions

Function	Description
TIME()	Extract time
TIMEDIFF()	Time difference
SEC_TO_TIME()	Seconds to time
TIME_TO_SEC()	Time to seconds

Examples

```
SELECT TIME(NOW());
```

```
SELECT TIMEDIFF('12:00:00', '10:30:00');
```

```
SELECT SEC_TO_TIME(3600);
```

```
SELECT TIME_TO_SEC('01:00:00');
```

12. Unix Timestamp Functions

Function	Description
UNIX_TIMESTAMP()	Convert to Unix timestamp
FROM_UNIXTIME()	Convert Unix timestamp to date

Examples

```
SELECT UNIX_TIMESTAMP();
```

```
SELECT UNIX_TIMESTAMP(NOW());
```

```
SELECT FROM_UNIXTIME(1750000000);
```

13. Time Zone Functions

View Current Time Zone

```
SELECT @@session.time_zone;
```

```
SELECT @@global.time_zone;
```

Set Time Zone

```
SET time_zone = '+05:30';
```

Convert Time Zone

```
SELECT CONVERT_TZ(  
    '2025-08-15 10:00:00',  
    '+00:00',  
    '+05:30'  
);
```

14. Important Date Functions

Function	Description
LAST_DAY()	Last day of month
MAKEDATE()	Create date
MAKETIME()	Create time
EXTRACT()	Extract part
TO_DAYS()	Days since year 0

Examples

```
SELECT LAST_DAY('2025-02-15');
```

```
SELECT MAKEDATE(2025, 100);
```

```
SELECT MAKETIME(10, 30, 45);
```

```
SELECT EXTRACT(YEAR FROM NOW());
```

```
SELECT TO_DAYS(NOW());
```

15. Working with NULL Dates

```
SELECT IFNULL(joining_date, CURDATE())  
FROM employee_logs;
```

16. Comparing Dates

Employees Joined After 2023

```
SELECT *  
FROM employee_logs  
WHERE joining_date > '2023-01-01';
```

Today's Records

```
SELECT *  
FROM employee_logs  
WHERE DATE(created_at) = CURDATE();
```

17. Sorting by Date

```
SELECT *  
FROM employee_logs  
ORDER BY created_at DESC;
```

18. Group By Month or Year

Count Employees by Year

```
SELECT
  YEAR(joining_date) AS joining_year,
  COUNT(*) AS total
FROM employee_logs
GROUP BY YEAR(joining_date);
```

Count by Month

```
SELECT
  MONTHNAME(joining_date) AS month_name,
  COUNT(*) AS total
FROM employee_logs
GROUP BY MONTH(joining_date);
```

19. Real-World Examples

Find Employees Joined in Last 30 Days

```
SELECT *
FROM employee_logs
WHERE joining_date >= CURDATE() - INTERVAL 30 DAY;
```

Find Age of Record

```
SELECT
  employee_name,
  TIMESTAMPDIFF(
    YEAR,
    joining_date,
    CURDATE()
  ) AS years_in_company
```

```
FROM employee_logs;
```

Convert User Input String to DATETIME

```
SELECT STR_TO_DATE(  
    '27-05-2026 10:45 PM',  
    '%d-%m-%Y %h:%i %p'  
);
```

20. Common Errors

Problem	Reason
Incorrect datetime value	Wrong format
NULL returned from STR_TO_DATE	Format mismatch
Invalid date	Non-existing date
Timezone issue	Wrong timezone settings

21. Best Practices

1. Use DATETIME for application timestamps.
 2. Use TIMESTAMP for automatic tracking.
 3. Always store dates in proper format.
 4. Avoid storing dates as VARCHAR.
 5. Use STR_TO_DATE() for user input conversion.
 6. Use DATE_FORMAT() only for presentation.
 7. Use UTC timezone in production systems.
 8. Index frequently filtered date columns.
-

22. Complete Practice Script

```
CREATE TABLE sales_orders (  
  order_id INT PRIMARY KEY AUTO_INCREMENT,  
  customer_name VARCHAR(100),  
  order_date DATETIME,  
  delivery_date DATE  
);
```

```
INSERT INTO sales_orders  
(customer_name, order_date, delivery_date)  
VALUES  
(  
  'Ramesh Iyer',  
  STR_TO_DATE(  
    '27-05-2026 10:30:45',  
    '%d-%m-%Y %H:%i:%s'  
  ),  
  DATE_ADD(CURDATE(), INTERVAL 7 DAY)  
);
```

```
SELECT  
  customer_name,  
  DATE_FORMAT(order_date, '%d-%m-%Y %h:%i %p') AS formatted_order_date,  
  DATEDIFF(delivery_date, CURDATE()) AS days_remaining  
FROM sales_orders;
```